A Flexible Reference-Insensitive Spatiotemporal Fusion Model for Remote Sensing Images Using Conditional Generative Adversarial Network

Zhenyu Tan, Meiling Gao, Xinghua Li, *Member, IEEE*, and Liangcun Jiang

***Abstract*— Due to the tradeoff between spatial and temporal resolutions of remote sensing images, spatiotemporal fusion models were proposed to synthesize the high spatiotemporal image series. Currently, spatiotemporal fusion models usually employ one coarse-resolution image acquired on a prediction date and at least another pair of coarse–fine resolution images close to the prediction time as references to derive the fine-resolution image on the prediction date. After years of development, the model accuracy has gained a certain improvement, but nearly, all the models require at least three image inputs and rigid time constraints must be applied to the references to guarantee the fusion accuracy. However, it is not always that easy to collect adequate data pairs for fine-resolution image series simulation in practice because of the bad weather condition or the time inconsistency between the coarse–fine resolution data sources, which causes some difficulties in the actual application. This article introduces the conditional generative adversarial network (CGAN) and switchable normalization technique into the spatiotemporal fusion problem and proposes a flexible deep network named the GAN-based SpatioTemporal Fusion Model (GAN-STFM) to reduce the number of model inputs and broke the time restriction on reference image selection. The GAN-STFM just needs a coarse-resolution image on the prediction date and another fine-resolution reference image at an arbitrary time in the same area for model inputs. As far as we know, this is the first spatiotemporal fusion model that requires only two images as model inputs and puts no restriction on the acquisition time of references. Even so, the GAN-STFM performs on par or better than other classical fusion models in the experiments. With this improvement, the data preparation for spatiotemporal fusion tends to be much easier than before, showing a promising perspective for practical applications.**

***Index Terms*— Conditional generative adversarial network (CGAN), convolutional neural network (CNN), data fusion, deep learning, generative adversarial network (GAN)-based SpatioTemporal Fusion Model (GAN-STFM), remote sensing, spatiotemporal.**

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I. Introduction

THE advances in sensor technologies promote the spatial and temporal resolutions of remotely sensed images continuously [1], benefiting massive practical remote-sensing-based applications, such as the monitoring and management of agriculture, environment, and resources, pushing its specialization and precision forward [2]. A simultaneous high spatial and high temporal resolution of remote sensing data, however, cannot be acquired due to the limitations of technologies and budgets [3], [4], and it partly confines the advanced applications of remote sensing. The spatiotemporal fusion model, served as a data postprocessing technique, is developed to lessen the hardware limitation. The spatiotemporal fusion process usually involves two kinds of remote sensing images. One data source equips with high spatial resolution and low temporal resolution (hereafter referred to as fine-resolution images). For instance, the Landsat-8 satellite acquires images at a spatial resolution of 30 m with a 16-day repeat cycle [5]. Another data source possesses a high temporal resolution and low spatial resolution (hereafter referred to as coarse-resolution images), such as the MODIS images. The MODIS sensor can capture daily Earth Observation data, but the spatial resolution can only reach 500 m for most of the spectral bands [6]. The spatiotemporal fusion technique combines this distinct information, by retrieving temporal ground changes from the coarse-resolution images and extracting detailed ground textures from the fine-resolution images, to synthesize new data with high spatial and high temporal resolutions simultaneously [3], [7], [8]. Fig. 1 shows the spatiotemporal fusion schematic. Usually, the coarse-resolution image on the prediction date t_1 and at least another coarse–fine resolution image pairs for reference at t_i $(i \neq 1)$ are involved in the spatiotemporal fusion. The time span between t_1 and t_i should be as small as possible, and there should be no significant ground feature changes during this time period. With spatiotemporal fusion models performed on long time-series data, high spatiotemporal image sequences can be generated and remote-sensing-based applications can be further extended to a fine-grained level. The whole fusion process can be formulated as (1), where F_{t_i} and C_{t_i} denote the fine-resolution image and coarse-resolution image on time t_i, respectively, and function f denotes the established spatiotemporal fusion algorithm, which is often quite complicated with numbers of rules

$$F_{t_1} = f(C_{t_0}, F_{t_0}, C_{t_1}, C_{t_2}, F_{t_2}). \tag{1}$$

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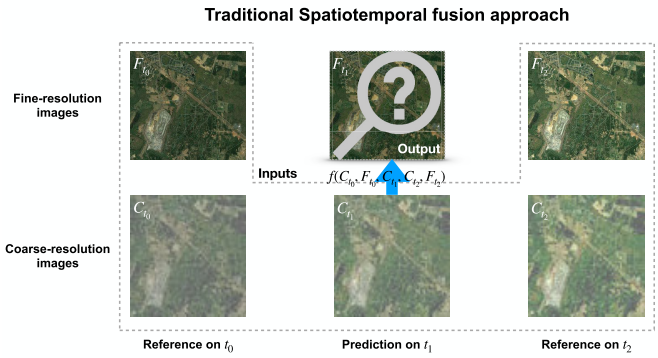


Fig. 1. Spatiotemporal fusion with coarse–fine resolution image pairs for references and coarse-resolution image on the prediction date (the image covered with question mark icon indicates the result to be predicted).

The study on spatiotemporal fusion of remote sensing images dates back to the 1990s [9]. After years of development, various spatiotemporal fusion models have been proposed [3], [9]. Generally, these models can be classified into four categories: 1) transformation-based; 2) reconstruction-based; 3) Bayesian-based; and 4) learning-based models [3], [10], [11]. Models based on transformation employ advanced mathematical transformation, such as wavelet transform, to map raw image pixels into an abstract space to perform the fusion [12], [13]. Reconstruction-based models can be further divided into weight function- and unmixing-based models [11]. A typical example of the weight function-based model is the spatial and temporal adaptive reflectance fusion model (STARFM) [7]. Many spatiotemporal fusion models are inspired by the STARFM model afterward, such as the spatial and temporal adaptive algorithm for mapping reflectance change (STAARCH) [14] and the enhanced STARFM (ESTARFM) [15], with the further improved accuracy. The unmixing-based algorithms establish the fusion model according to the linear unmixing theory, such as the spatial–temporal data fusion approach (STDFA) [16]. Some research works combine the weighted-function-based idea with the unmixing-based approach to build more advanced fusion models, such as the spatial and temporal reflectance unmixing model (STRUM) [17] and the Flexible Spatiotemporal DAta Fusion (FSDAF) [18]. The Bayesian-based fusion models are established based on the Bayesian statistics, trying to integrate the spatial–temporal–spectral fusion into a unified framework [4], [19]. The learning-based models are built by learning rules from archived data. It can be roughly categorized into the sparse-representation-based, such as SParse-representation-based SpatioTemporal reflectance Fusion Model (SPSTFM) [20], and the deep-learning-based models, such as spatiotemporal fusion using deep convolutional neural networks (STFDCNN) [21], Deep Convolutional SpatioTemporal Fusion Network (DCSTFN) [8], enhanced DCSTFN (EDCSTFN) [10], and a two-stream convolutional neural network (StfNet) [22].

The fusion models based on deep learning technology can learn implicit complex rules and relations from massive archived data sets with modern computing hardware acceleration, boosting the efficiency and accuracy of the computer modeling considerably [23]. By introducing the deep convolutional neural network (CNN) into the remote sensing image fusion problem, the model accuracy is significantly promoted, showing a promising research direction [8], [10], [21], [22]. However, many spatiotemporal fusion models are trained to perform prediction on the ideal data set, neglecting the difficulties to collect the satisfactory data for some study areas. Currently, nearly, all the existing models require at least three input images, and rigid constraints must be applied to reference image selection to guarantee fusion accuracy. However, it is not always that easy to collect adequate data pairs for a continuous period of time in practice because of the bad weather condition, data missing in the study area, or the time inconsistency between the coarse–fine resolution data sources [24]. Thus, the strong time dependence for reference images in the spatiotemporal fusion problem needs to be alleviated. To this end, this article utilizes the thoughts of generative adversarial network (GAN) and normalization techniques to ameliorate the aforementioned problem. A novel architecture named GAN-based SpatioTemporal Fusion Model (GAN-STFM) is proposed in this article and comparisons with some existing spatiotemporal fusion models later demonstrate its effectiveness. The contributions of this essay are twofold. First, the GAN architecture is introduced into the spatiotemporal fusion domain to refine the model output. Second, unlike the existing spatiotemporal fusion models, the restriction on the acquisition time of reference images is eliminated. With the GAN-STFM, preparing input data for the fusion model can be less painstaking. We believe that the reference-insensitive fusion approach can give a reference to future research and have a promising perspective for practical applications.

The rest of this article is organized as follows. The basics of GAN and the detailed architecture of the proposed GAN-STFM is introduced in Section II. Section III dives into the details of the analysis and comparison of the model results. Conclusion and future work are summarized in Section IV.

II. Materials and Methods

A. Generative Adversarial Network

The GAN was first proposed by Goodfellow *et al.* [25], and then, an explosion of GAN-based applications is occurred, such as image generation [26], super-resolution [27], denoising [28], inpainting [29], and image-to-image translation [30], showing its powerful modeling capability. GAN, as the name suggests, is a kind of generative model. It is trained in the competing process between a generator and a discriminator, where the discriminator is designed to distinguish the generator outputs between training samples, and meanwhile, the generator makes efforts to generate fake data, which resembles the same distribution of the real samples, according to the feedback from the discriminator. Through this iterative competitive training, they both evolve to be stronger and the whole system achieves a state of equilibrium in the end. Finally, the generator can create fake outputs almost like the real samples, and the discriminator cannot tell differences between the real and fake ones. In essence, GAN can be regarded as a two-person zero-sum game.

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Mathematically, the generator G takes some noisy data $\boldsymbol{z}$ as inputs, learns a complex mapping function over real training samples $\boldsymbol{x}$, and tries to map the distribution of noisy data $p_z(z)$ to the distribution of real data $P_{\text{data}}(\boldsymbol{x})$. Meanwhile, the discriminator D is served as a binary classifier to distinguish the generated fake data $G(z)$ and real samples $\boldsymbol{x}$. In other words, the generator aims to minimize the distribution distance between $G(z)$ and $\boldsymbol{x}$, while the goal of the discriminator is to maximize the distribution between them. Therefore, the vanilla GAN employs a minimax loss function, formulated as (2), to train the generator and discriminator simultaneously [25]

$$\min_G \max_D V_{\text{GAN}}(D, G) = \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))]. \quad (2)$$

The vanilla GAN loss function measures the distribution distance via the Jensen–Shannon (JS) divergence under the hood. Without additional constraints on the solution space, the vanilla GAN suffers weak stability for model training in practice [31]. Many studies afterward dedicate to explore possible solutions to impose restrictions on vanilla GAN to improve its performance. For instance, the conditional GAN (CGAN) [32] feeds an additional data $\boldsymbol{y}$ into both the generator and discriminator as a conditional label to drive the network to search for the optimum solution along the expected direction, solving the problem that the vanilla GAN has no control over the generated data modes. The CGAN loss function is denoted as (3). The least squares GAN (LSGAN) [33] is another GAN variant with small modification on the loss function to improve the GAN stability. Practices show the LSGAN can produce more high-quality results compared with the vanilla GAN, especially for image-related tasks [33], [34]. Hence, it is adopted for the GAN-STFM training instead of the original Minimax loss. The LSGAN loss function is formulated as (4), where a, b, and c are constants, satisfying the conditions of $b - c = 1$ and $b - a = 2$ [33]

$$\min_G \max_D V_{\text{CGAN}}(D, G) = \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x} \mid \boldsymbol{y})] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z \mid \boldsymbol{y})))] \quad (3)$$

$$\min_D V_{\text{LSGAN}}(D) = \frac{1}{2}\mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}\left[(D(\boldsymbol{x}) - b)^2\right] + \frac{1}{2}\mathbb{E}_{z\sim p_z(z)}\left[(D(G(z)) - a)^2\right]$$

$$\min_G V_{\text{LSGAN}}(G) = \frac{1}{2}\mathbb{E}_{z\sim p_z(z)}\left[(D(G(z)) - c)^2\right]. \quad (4)$$

B. GAN-STFM Architecture

1) Model Design: Theoretically, an image can be decomposed into coarse approximation and fine details. In the spatiotemporal fusion problem, the coarse-resolution image on the prediction date can be treated as the approximation part, and the ground details on the prediction date need to be conjectured from a fine-resolution reference image. The main idea for the reference-insensitive model is that a deep model can be utilized to extract common high-pass features from the fine-resolution reference images, and then, the expected fusion result can be compounded with the approximation on prediction date and the common details from an arbitrary fine-resolution reference image under the assumption that little shape changes of the ground features occurred during the reference and prediction time period in the study area. Based on this idea, this article introduces the CGAN and normalization technique into the remote sensing spatiotemporal fusion problem and proposes a novel model with great flexibility named GAN-STFM to alleviate the strong time dependence between reference and prediction images in the existing fusion models. Fig. 2 shows the reference-insensitive spatiotemporal fusion approach conceptually. Mathematically, the reference-insensitive model can be formulated as (5), where F_t and C_t denote the synthesized fine-resolution image and observed coarse-resolution image on the prediction date time t, respectively, F_{t^*} denotes an arbitrary fine-resolution image on time t^* ($t^* \neq t$) for reference, and function f denotes the learned mapping relation between inputs and output. By comparing with (1), it can be seen that the GAN-STFM model is much simpler formally than the traditional spatiotemporal fusion approach

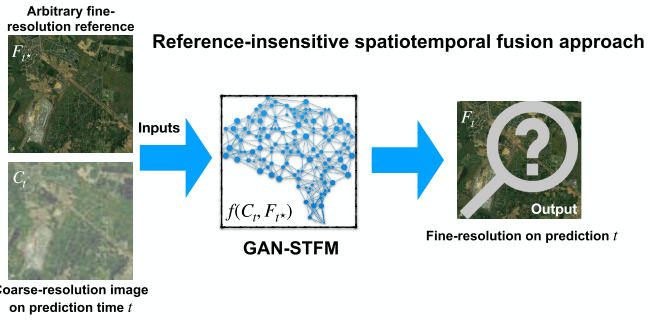


Fig. 2. Reference-insensitive spatiotemporal fusion approach with only one coarse-resolution image on the prediction date and another arbitrary fine-resolution image for reference (the reference image is selected randomly from the fine-resolution data set in the training process).

$$F_t = f(C_t, F_{t^*}). \quad (5)$$

The proposed GAN-STFM network, of course, needs a generator and discriminator, shown in Fig. 3, and they are both constructed based on the ResNet [35] building blocks. With ResNet, the relation between input(s) and output(s) is not established directly, instead, the residuals between input(s) and output(s) are learned with skip connections. Due to the residual architecture, the vanishing and exploding gradient problems in very deep networks are significantly mitigated. The basic ResNet blocks used to compose the GAN model are shown in Fig. 4. Two different building blocks are involved in the generator because, conceptually, the generator can be divided into an encoder and a decoder. For the encoder, the main branch is used to extract the common high-pass features, and then, the intermediate fine features are generated by adding the common high-pass features to the image-specific approximation from the lateral branch. Fig. 4(a) shows this process (GEncoder-ResBlock) that an arbitrary fine-resolution image for reference is fed into the main branch, whereas a coarse-resolution image on the prediction date goes through the lateral branch. Then, the merged intermediated features

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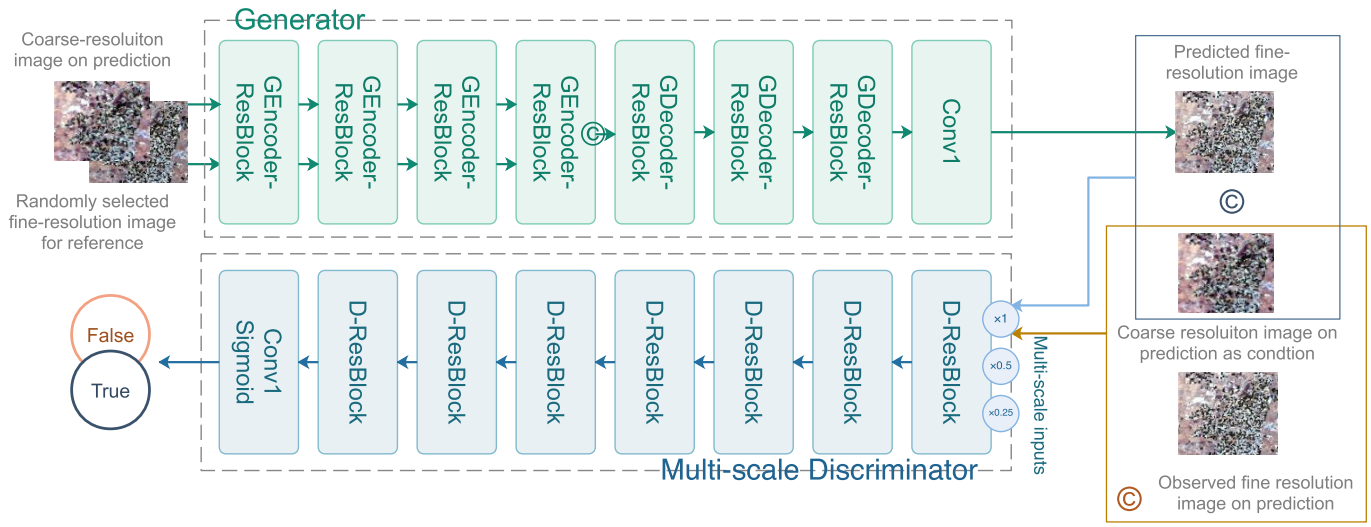


Fig. 3. Whole architecture of the GAN-STFM model for spatiotemporal fusion (GEncoder-ResBlock and GDecoder-ResBlock represent the residual blocks for the generator encoder and decoder, respectively. The D-ResBlock is the residual block used in the discriminator. The details of these basic building blocks are shown in Fig. 4. The inputs of the generator are a coarse image on the prediction date and another fine-resolution reference image at an arbitrary time in the same area for model inputs. The inputs of the discriminator include two groups: one is the predicted fine-resolution image from the generator and the coarse-resolution image on prediction and another is the observed fine-resolution image and the coarse-resolution image on prediction. The inputs of the original discriminator are downsampled to three levels with the factor of 1, 0.5, and 0.25 to train the multiscale discriminator. The © symbol denotes the concatenation of raw images or feature maps. The arrows denote the data flow direction in the network).

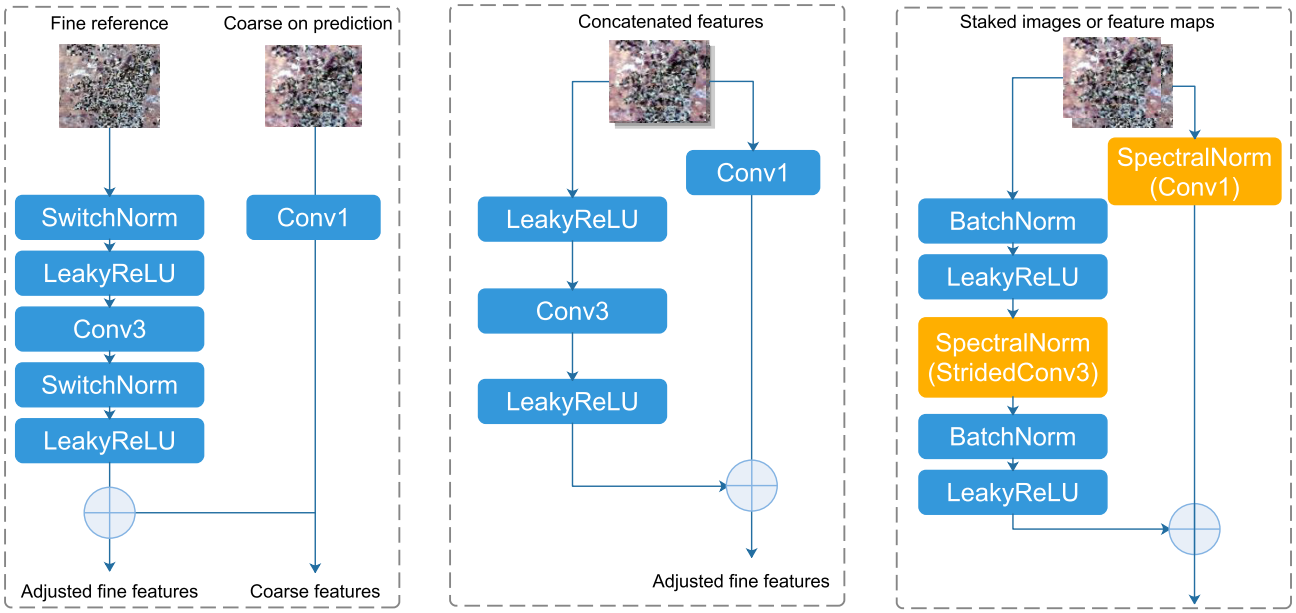


Fig. 4. Basic residual building blocks for the generator and the discriminator (Conv1 and Conv3 denote the convolution operations with the kernel size of 1 and 3, respectively, and StridedConv3 represents the convolution operation with the kernel size of 3×3 and the stride of 2. The $\bigoplus$ denotes the elementwise addition of feature maps. SwitchNorm is short for the switchable normalization, and BatchNorm is the batch normalization, applied to the feature maps. Spectral normalization is applied to the convolution parameters in orange for the Discriminator). (a) Residual block for the generator encoder (GEncoder-ResBlock). (b) Residual block for the generator decoder (GDecoder-ResBlock). (c) Residual block for the discriminator (D-ResBlock).

and the coarse approximation features are sent to the next adjacent blocks separately. This separation aims to maintain the coarse spectral information as much as possible in the forward propagation. At the end of the encoder, the adjusted fine features and coarse features are extracted in an abstract high-level space and then concatenated and delivered to the decoder. The concatenated features go through the main and lateral branches at the same time in the decoder. The intermediate features are added and then sent to the next residual block. The building block (GDecoder-ResBlock) adopted in the decoder of the generator is shown in Fig. 4(b), which is the same as the one used in the original ResNet architecture.

The generator is constructed with the stacked residual blocks, as shown in Fig. 3. The first four blocks in the generator formed the encoder with GEncoder-ResBlocks, and the decoder is composed of the rest four with GDecoder-ResBlocks. Another difference between the generator encoder and decoder architectures lies in the switchable normalization,

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as shown in Fig. 4. It is utilized in the generator encoder to compel the model to filter out sample-specific information and just extract the common high-pass details. The switchable normalization is a learnable normalization method driven by tasks and data, denoted as (6) [36], where h_{ncij} and $\hat{h}_{ncij}$ are the feature values with coordinates (n, c, i, j) before and after the normalization; Ω represents a class of statistics estimated with different normalization manners, including batch normalization [37], layer normalization [38], and instance normalization [39]; μ and σ denote the mean and standard deviation, respectively; β and γ are the shift and scale parameters, respectively; and w_k and w'_k are the learnable weights to be applied to the statistics estimated in different ways. It can be seen from (6) that the switchable normalization unifies the existing three feature normalization methods, suitable for the case that common features are expected to extract. The activation after the normalization layer is leaky rectified linear unit (LeakyReLU) function [40], which is extensively employed in CNN models. The Conv3 layer in the main branch, as shown in Fig. 4, denotes the plain convolution operation with the kernel size of 3×3. Conv1 used in the lateral branch is to adjust the channels of feature maps to match the lateral branch output without introducing too many parameters

$$\hat{h}_{ncij} = \gamma \frac{h_{ncij} - \Sigma_{k\in\Omega} w_k \mu_k}{\sqrt{\Sigma_{k\in\Omega} w'_k \sigma_k^2 + \epsilon}} + \beta. \tag{6}$$

The discriminator is an ordinary binary classifier composed of several D-ResBlocks. Batch normalization is used in each residual block, shown in Fig. 4(c), to generalize the model. StridedConv3 denotes the convolution operation with the kernel size of 3×3 and the stride of 2 to reduce the feature size. The distinctive structure of the discriminator is that the spectral normalization is employed to stabilize the adversarial process in the GAN training. Spectral normalization is applied to the weights of convolution operations, instead of intermediate feature maps, to guarantee the Lipschitz continuity of the learned implicit function [41]. The inputs of the discriminator are the coarse-resolution image on the prediction date and the fine-resolution ground truth or the predicted image. The coarse-resolution image on the prediction date serves as the condition to guide the training process. With the last sigmoid activation, the discriminator maps the input images to the probabilities of the false and true groups. Beyond all that, a multiscale discriminator introduced in [42] is employed in the GAN-STFM to enhance the output image quaintly. The conditioned inputs of the discriminator are downsampled to two different scales with the factors of 0.5 and 0.25 and used to train three discriminators with the same architecture at different scales, as shown in Fig. 3.

2) Loss Function: The LSGAN loss function is selectively used in the GAN-STFM model with additional constraints on image content. The calculation of losses of the discriminator follows the same as the corresponding part in the LSGAN, whereas the losses of the generator are compounded of the corresponding LSGAN part ($\mathcal{L}_{\text{GAN}}$) and the constraints on the image itself ($\mathcal{L}_I$). The losses for image itself include the feature loss ($\mathcal{L}_{\text{Feature}}$), spectral angle loss ($\mathcal{L}_{\text{Spectrum}}$), as well as the vision loss ($\mathcal{L}_{\text{Vision}}$) [10]. Thus, the total losses for the generator can be formulated as (7), where α, β, and λ are the weight coefficients

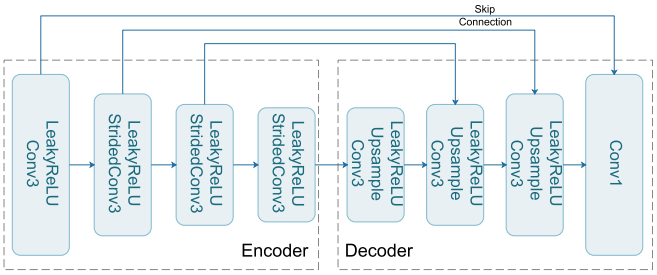


Fig. 5. Pretrained AutoEncoder for feature loss calculation (the connection line with arrow stands for the skip connection, and it can concatenate lower level features with higher level features).

$$\begin{aligned}\min_G V(G) &= \mathcal{L}_I + \lambda \mathcal{L}_{\text{GAN}} \\ &= \mathcal{L}_{\text{Feature}} + \alpha \mathcal{L}_{\text{Spectrum}} + \beta \mathcal{L}_{\text{Vision}} + \lambda \mathcal{L}_{\text{GAN}}. \end{aligned} \tag{7}$$

The feature loss is used to capture the perceptual differences to generate high-quality images. It can be calculated with the help of a pertained AutoEncoder, reducing the differences in an abstract high feature level. The pretrained AutoEncoder also consists of an encoder and decoder, and it is constructed as a fully convolutional network with skip connections forming a classical "hourglass" architecture, as shown in Fig. 5. The StridedConv3 layer in the encoder with the stride of 2 is used to shrink the feature size by half, whereas the Conv3 layer in the decoder, as well as the following Upsample layer, is to expand the feature size twice. The last layer Conv1 is used to restore feature maps to raw images. The input and output of the AutoEncoder are the same fine-resolution image. The encoder maps raw images to high-dimensional feature space and the decoder restores these abstract features to the original image space. The feature losses in the GAN-STFM are calculated using the mean squared error (MSE) function based on the output features of the encoder, shown as (8), where $\boldsymbol{x}$ and $\boldsymbol{y}$ denote the ground truth and prediction, respectively, and $f_{\text{Encoder}}(x)$ and $f_{\text{Encoder}}(y)$ are the outputs of the pretrained encoder. The spectral angle loss is proposed in this article to diminish and control the spectral distortion among different bands using the cosine similarity between the prediction and ground-truth features. It is calculated according to (9), where $\boldsymbol{I}$ denotes the tensor of ones. The vision loss is used to retain more detailed ground textures from the perspective of computer vision. It can be computed using the multiscale structural similarity (MS-SSIM) [43] to guarantee a satisfactory visual effect, denoted as (10), where $l_i(\boldsymbol{x}, \boldsymbol{y})$, $l_i(\boldsymbol{x}, \boldsymbol{y})$, and $l_i(\boldsymbol{x}, \boldsymbol{y})$ are the quantified luminance, contrast, and structure of the image pairs at the ith scale, respectively, and α_i, β_i, and γ_i are the corresponding weights coefficients

$$\mathcal{L}_{\text{Feature}} = \frac{1}{N} \sum_{i=1}^{N} \left[f_{\text{Encoder}}(\boldsymbol{x}) - f_{\text{Encoder}}(\boldsymbol{y}) \right]^2 \tag{8}$$

$$\mathcal{L}_{\text{Spectrum}} = \boldsymbol{I} - \frac{f_{\text{Encoder}}(\boldsymbol{x}) \cdot f_{\text{Encoder}}(\boldsymbol{y})}{\| f_{\text{Encoder}}(\boldsymbol{x}) \| \| f_{\text{Encoder}}(\boldsymbol{y}) \|} \tag{9}$$

$$\mathcal{L}_{\text{Vision}} = \boldsymbol{I} - \left[l_M(\boldsymbol{x}, \boldsymbol{y}) \right]^{\alpha_M} \cdot \prod_{i=1}^{M} \left[c_i(\boldsymbol{x}, \boldsymbol{y}) \right]^{\beta_i} \left[s_i(\boldsymbol{x}, \boldsymbol{y}) \right]^{\gamma_i}. \tag{10}$$

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III. Experiments and Results

A. Study Areas and Data Sets

Two open-source data sets from the Coleambally Irrigation Area (CIA)[1] and the Lower Gwydir Catchment (LGC)[2] study areas are used in the following experiments [44]. The CIA study area is located in southern New South Wales, Australia for rice plants with a modern irrigation system. It is known for the significant temporal dynamics regarding the crop phenology over a growing season. There are 17 cloud-free MODIS-Landsat data pairs available from 2001 to 2002 in the CIA study area. Each image is composed of six spectral bands. The MODIS and Landsat images are both interpolated to the size of 2040×1720 at 25-m resolution. The LGC study area is located in northern New South Wales, Australia, and 14 cloud-free data pairs from 2004 to 2005 are collected with the interpolated image size of 2720×3200 for spatiotemporal fusion. This area can also be considered temporally dynamic because of the flooding event. All the data have gone through the standard preprocessing process and can be directly used for the following fusion models.

B. Experiments Settings

Three goals are expected to be realized in the following experiments. First, the ablation experiment was performed by removing some building components from the GAN-STFM architecture to gain insight into its effects. Second, the fusion accuracy of GAN-STFM was evaluated by comparing it with some existing spatiotemporal fusion models. Third, the insensitivity of the GAN-STFM to reference images was demonstrated by using different reference images to perform predictions on a specific date. The GAN-STFM was compared with the following fusion algorithms, including the STARM, FSDAF, and EDCSTFN. The STARFM algorithm is a classical weighted reconstruction-based fusion model. The FSDAF is an unmixing-based reconstruction fusion model. The EDCSTFN is a deep-learning-based fusion model. All three models require at least a coarse-resolution image on the prediction date and a pair of coarse–fine resolution images for reference. In contrast, the GAN-STFM only needs a coarse-resolution image on the prediction date and an arbitrary fine-resolution reference image. To train the EDCSTFN model, each MODIS-Landsat data pair needs to be grouped with another data pair, and the data acquisition dates should be close enough in a group. Namely, the acquisition time of the reference images is the closest to the prediction date in all the grouped data pairs. The data group rule is also applied to the STARM and FSDAF models. For the GAN-STFM model, prediction and reference data pairs do not need to be grouped because the fine-resolution reference is randomly selected from the Landsat data set for a specifics study area. This means that the reference image for a specific prediction date is not the same for every training iteration and the GAN-STFM model can be trained in this way to learn the reference-insensitive high-pass features. After grouping all the data sets, ten data groups in 2002 were selected for the CIA study area, and five groups in 2001 were used to validate the accuracy. Validation data did not involve in the training process. For the LGC study area, eight data groups in 2004 were selected to train models and four groups in 2005 for validation.

TABLE I
Experiment Hardware and Software Environment

Hardware	RAM	CPU	GPU
	125G	2×Intel Xeon Gold 5218R	2×RTX 3090
Software	Python	CUDA	PyTorch
	3.7	11.0	1.7

TABLE II
Information on the Model Training

Models	Parameters	Time/Batch (s)
EDCSTFN	284134	1.096
GAN-STFM	578130 (Generator) 3665959 (Discriminator)	1.492

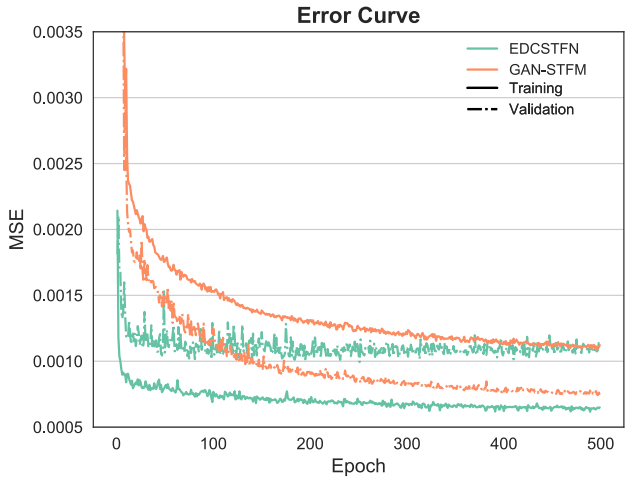
The deep-leaning-based models are both implemented with Python programming language using the PyTorch deep learning framework [45]. To contribute to the geoscience community, the source code is unveiled in the GitHub[3] platform, available for public testing and use. When performing the model training, each image is subdivided into small patches (256×256) to feed into the networks considering the massive computer memory consumption. There are eight basic building blocks in the instantiated generator, as shown in Fig. 3, the output channels for each block are set to 16, 32, 64, 128, 64, 32, 16, and 6, empirically. The output channels of the last layer equal to the number of image bands. For the discriminator, eight building blocks are needed corresponding to the 256×256 image patch size. The output channels for each block are empirically set to 12, 32, 32, 64, 64, 128, 128, 256, and 256. The input channel of the first layer should be twice as the number of spectral bands because the fine-resolution images of prediction or ground truth and the conditional coarse-resolution image on the prediction date are stacked to feed into the discriminator. The initial learning rate for training is set to 2×10^{-4} for the GAN-STFM and 1×10^{-3} for the EDCSTFN. The batch size for each epoch is set to 32 patches with the stride size of 200 and 500 epochs that are run in the experiments. The coefficients in (7) are set as follows: $\alpha = \beta = 1$ and $\lambda = 0.01$. The hyperparameters of the model can be tweaked according to actual situations, such as the hardware specifications and experiment data. Our experiment environment were listed in Table I.

C. Training the Models

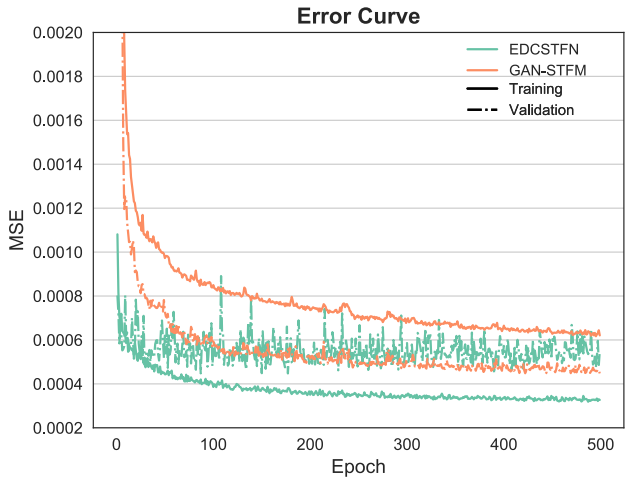
The deep-learning-based models were trained with the acceleration of graphics processing units (GPUs). The model parameters and running time are summarized in Table II based on the aforementioned settings. Surprisingly, the EDCSTFN with rather fewer parameters takes just slightly less time than the GAN-STFM with much more complex architecture.

[1] http://dx.doi.org/10.4225/08/5111AC0BF1229CIA
[2] http://dx.doi.org/10.4225/08/5111AD2B7FEE6LGC
[3] https://github.com/theonegis/ganstfm

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(a)



(b)

Fig. 6. Error variation during models training on the training and validation data sets. (a) Error curves for the CIA data set. (b) Error curves for the LGC data set.

This is because loading groups of data from disk into GPU memory is quite time-consuming, and GAN-STFM only needs two thirds of input images compared with EDCSTFN, which can save some time when the computing performance of GPUs is not a system bottleneck in the model training. The MSE variation during deep models training on validation data sets is presented in Fig. 6. It comes as no surprise that the EDCSTFN is converged more quickly than the GAN-STFM. This is because the EDCSTFN model requires three input images and the time during the reference and prediction date is quite close, and hence, there are a few significant gourd changes during a short time period. The relation between inputs and output is relatively not that complicated compared with the GAN-STFM. The GAN-STFM model is designed to accept two input images only, and the fine-resolution reference is randomly selected in the training. In this case, the relation between inputs and output is relatively complex than the EDCSTFN model because the GAN-STFM algorithm needs to extract common high-pass features from massive fine-resolution references instead of sample-specific features. The training errors of EDCSTFN from Fig. 6 are smaller than the GAN-STFM model in both the CIA and LGC study areas, while the opposite happened regarding the validation errors. This confirms that the EDCSTFN is easy to train than the GAN-STFM model, and moreover, the GAN-STFM shows a higher generalization ability.

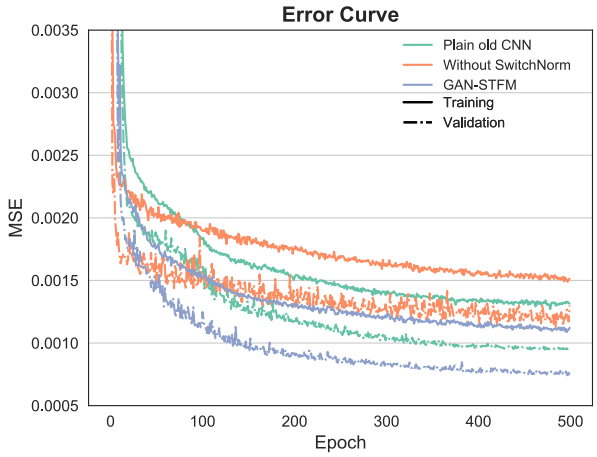
D. Comparison and Results

To evaluate the performances with different fusion models objectively, several quantitative indices are chosen to perform the comparisons, including mean absolute error (MAE), root-mean-square error (RMSE), spectral angle mapper (SAM) [46], and structural similarity index (SSIM) [47]. The MAE and RMSE are often used to quantify fusion errors, and a small value indicates better fusion accuracy. The SAM evaluates fusion models concerning the spectral distortion, and a small value suggests a high spectrum match. The SSIM reveals the synthesized image quality, and a desirable fused image usually comes with high value. The following represents the experiment results from three perspectives corresponding to the aforementioned three goals.

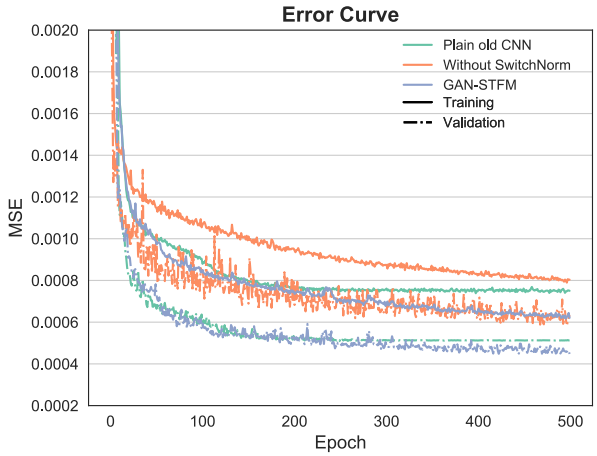
1) Ablation Study: Compared with the existing deep-learning-based fusion models, first of all, the switchable normalization is utilized to reduce the dependence of fine reference images in the fusion process. Then, the CGAN-based architecture is employed in the GAN-STFM to improve the generated image quality. Therefore, the two components were selectively removed to understand the contribution to the overall fusion model in this section. To be specific, the plain old CNN model without discriminator, the GAN-based model without switchable normalization in the generator, and the GAN-STFM model were trained. Fig. 7 shows the error variation during training for the ablation study. Generally, the GAN-STFM has the lowest errors in both two study areas on the training and validation data sets. The model without switchable normalization has the highest errors and the errors fluctuate on the validation data set in the training process, showing the effectiveness of switchable normalization in the reference-insensitive model. The plain CNN model performs quite steady than the one without switchable normalization, but still not so good as the GAN-based model. Tables III and IV list the quantitative evaluation results on the validation data sets with the three architectures. The bold numbers highlight the best results. This, again, confirms the fact that the GAN architecture and switchable normalization play a significant part in the GAN-STFM fusion model.

2) Model Accuracy Comparison: Five data groups in 2001 for the CIA area and four data groups in 2005 for the LGC are tested to evaluate the model accuracy. The quantitative comparisons are listed in Tables V and VI. Overall, the deep-learning-based models outperform traditional fusion methods observably. The EDCSTFN and GAN-STFM are evenly matched. Specifically, one performs better in some data groups regarding some metrics, whereas another produces better results in other data groups regarding other metrics. Considering that the GAN-STFM only needs two input images and the reference image can be chosen with fewer constraints, the GAN-STFM model shows more

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(a)



(b)

Fig. 7. Error variation on three different model architectures (the plain old CNN model without discriminator, the GAN-based model without switchable normalization in the generator, and the GAN-STFM model). (a) Error curves for the CIA data set. (b) Error curves for the LGC data set.

TABLE III

QUANTITATIVE EVALUATION RESULTS FOR THE CIA AREA ON DIFFERENT DEEP MODEL ARCHITECTURES IN 2001

Prediction Date	Model	MAE	RMSE	SAM	SSIM
Oct. 17th	Plain old CNN	0.0188	0.0327	0.1901	0.8550
	Wihout SiwthNorm	0.0213	0.0369	0.2041	0.8361
	GAN-STFM	**0.0164**	**0.0294**	**0.1646**	**0.8817**
Nov. 2nd	Plain old CNN	0.0196	0.0324	0.1896	0.8440
	Wihout SiwthNorm	0.0191	0.0324	0.1805	0.8381
	GAN-STFM	**0.0165**	**0.0290**	**0.1658**	**0.8664**
Nov.9th	Plain old CNN	0.0167	0.0298	0.2024	0.8662
	Wihout SiwthNorm	0.0168	0.0312	0.2050	0.8598
	GAN-STFM	**0.0152**	**0.0285**	**0.1921**	**0.8801**
Nov. 25th	Plain old CNN	0.0154	0.0260	0.1511	0.8825
	Wihout SiwthNorm	0.0174	0.0274	0.1501	0.8718
	GAN-STFM	**0.0131**	**0.0232**	**0.1302**	**0.9042**
Dec. 4th	Plain old CNN	0.0214	0.0380	0.1771	0.8551
	Wihout SiwthNorm	0.0198	**0.0310**	**0.1482**	0.8483
	GAN-STFM	**0.0197**	0.0370	**0.1623**	**0.8738**

TABLE IV

QUANTITATIVE EVALUATION RESULTS FOR THE LGC AREA ON DIFFERENT DEEP MODEL ARCHITECTURES IN 2005

Prediction Date	Model	MAE	RMSE	SAM	SSIM
Jan. 13th	Plain old CNN	0.0162	0.0237	0.1368	0.8917
	Wihout SwitchNorm	0.0165	0.0244	0.1345	0.8867
	GAN-STFM	**0.0153**	**0.0226**	**0.1315**	**0.9007**
Jan. 29th	Plain old CNN	0.0175	0.0261	0.1479	0.8873
	Wihout SwitchNorm	0.0171	0.0267	0.1499	0.8786
	GAN-STFM	**0.0166**	**0.0246**	**0.1367**	**0.9011**
Feb. 14th	Plain old CNN	0.0140	0.0204	0.1207	0.9138
	Wihout SwitchNorm	0.0147	0.0228	0.1267	0.9036
	GAN-STFM	**0.0126**	**0.0189**	**0.1083**	**0.9269**
Mar. 2nd	Plain old CNN	0.0149	0.0219	0.1197	0.9128
	Wihout SwitchNorm	0.0149	0.0222	0.1190	0.9081
	GAN-STFM	**0.0132**	**0.0194**	**0.1065**	**0.9275**

TABLE V

QUANTITATIVE EVALUATION RESULTS FOR THE CIA AREA WITH VARIOUS SPATIOTEMPORAL FUSION METHODS IN 2001

Prediction Date	**Model**	**MAE**	**RMSE**	**SAM**	**SSIM**
Oct. 17th	STARFM	0.0502	0.1952	1.0840	0.8077
	FSDAF	0.0218	0.0427	0.2597	0.8405
	EDCSTFN	0.0203	0.0326	0.1767	0.8735
	GAN-STFM	**0.0164**	**0.0293**	**0.1645**	**0.8817**
Nov. 2nd	STARFM	0.0804	0.2722	1.2679	0.7699
	FSDAF	0.0267	0.0549	0.3176	0.8088
	EDCSTFN	0.0192	0.0317	0.1897	0.8626
	GAN-STFM	**0.0164**	**0.0290**	**0.1658**	**0.8663**
Nov. 9th	STARFM	0.0224	0.0808	0.3704	0.8261
	FSDAF	0.0177	0.0321	0.2183	0.8639
	EDCSTFN	0.0159	0.0290	0.1949	0.8768
	GAN-STFM	**0.0151**	**0.0284**	**0.1921**	**0.8801**
Nov. 25th	STARFM	0.0252	0.1006	0.6010	0.8351
	FSDAF	0.0181	0.0315	0.1849	0.8749
	EDCSTFN	0.0174	0.0260	0.1376	0.8954
	GAN-STFM	**0.0131**	**0.0231**	**0.1301**	**0.9042**
Dec. 4th	STARFM	0.0831	0.2829	1.1906	0.7858
	FSDAF	0.0287	0.0600	0.2860	0.8215
	EDCSTFN	0.0261	**0.0345**	**0.1410**	0.8640
	GAN-STFM	**0.0197**	0.0369	0.1623	**0.8737**

TABLE VI

QUANTITATIVE EVALUATION RESULTS FOR THE LGC AREA WITH VARIOUS SPATIOTEMPORAL FUSION METHODS IN 2005

Prediction Date	**Model**	**MAE**	**RMSE**	**SAM**	**SSIM**
Jan. 13th	STARFM	0.0174	0.0342	0.2166	0.8590
	FSDAF	0.0172	0.0260	0.1305	0.8867
	EDCSTFN	**0.0145**	**0.0217**	**0.1242**	**0.9037**
	GAN-STFM	0.0153	0.0225	0.1314	0.9006
Jan. 29th	STARFM	0.0187	0.0448	0.2576	0.8498
	FSDAF	0.0163	0.0251	0.1336	0.8925
	EDCSTFN	**0.0154**	**0.0233**	**0.1267**	**0.9078**
	GAN-STFM	0.0166	0.0246	0.1366	0.9010
Feb. 14th	STARFM	0.0143	0.0319	0.1977	0.8800
	FSDAF	0.0135	0.0227	0.1312	0.9210
	EDCSTFN	**0.0123**	**0.0188**	0.1102	0.9244
	GAN-STFM	0.0125	**0.0188**	**0.1082**	**0.9269**
Mar. 2nd	STARFM	0.0141	0.0329	0.1980	0.8859
	FSDAF	0.0132	0.0224	0.1223	0.9221
	EDCSTFN	0.0144	0.0207	0.1068	0.9222
	GAN-STFM	**0.0131**	**0.0194**	**0.1064**	**0.9275**

flexibility and has a quite promising prospect for practical data production.

The synthesized fusion results for the CIA area on November 9, 2001 are shown in Fig. 8. The color images composed of the bands 5, 4, and 3 are chosen for the visual demonstration. The output of the STARFM algorithm contains

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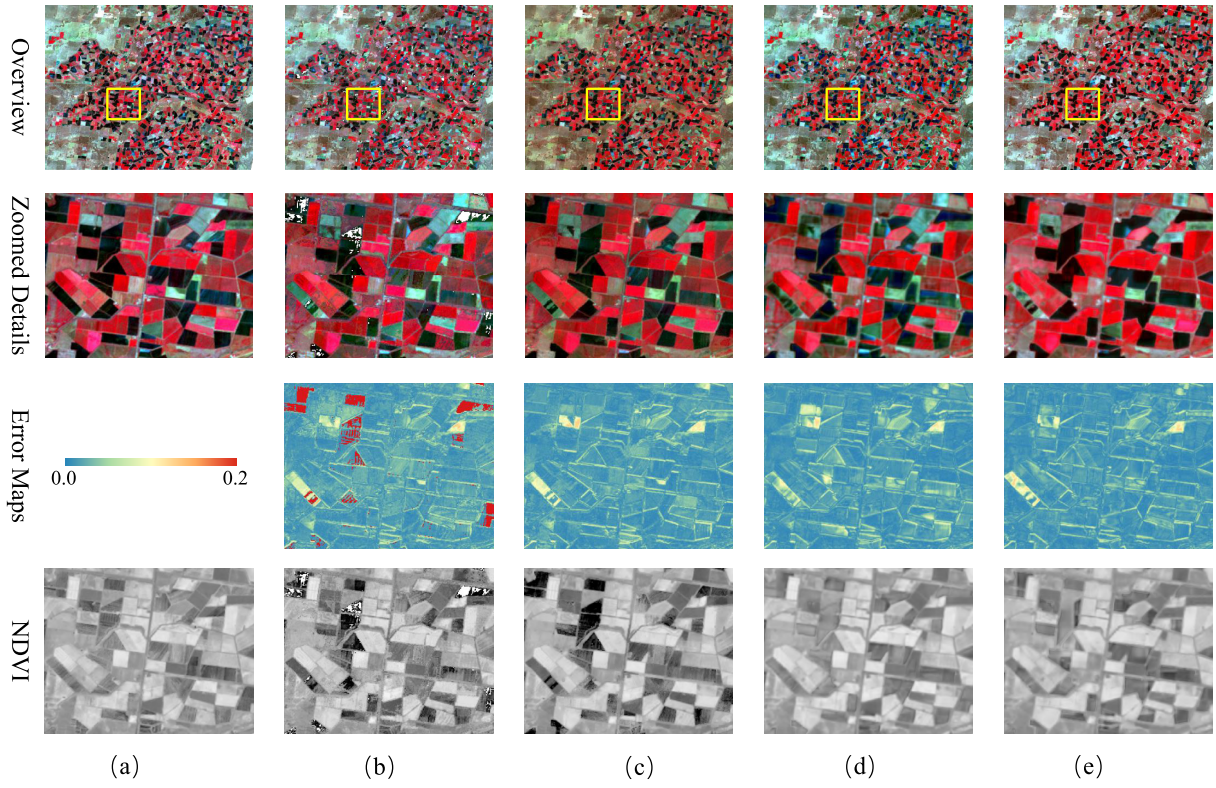


Fig. 8. Fusion results for the CIA area on November 9, 2001 with different spatiotemporal fusion methods (the last three rows correspond to the region in the first row marked with yellow rectangles). (a) Ground Truth. (b) STARFM. (c) FSDAF. (d) EDCSTFN. (e) GAN-STFM.

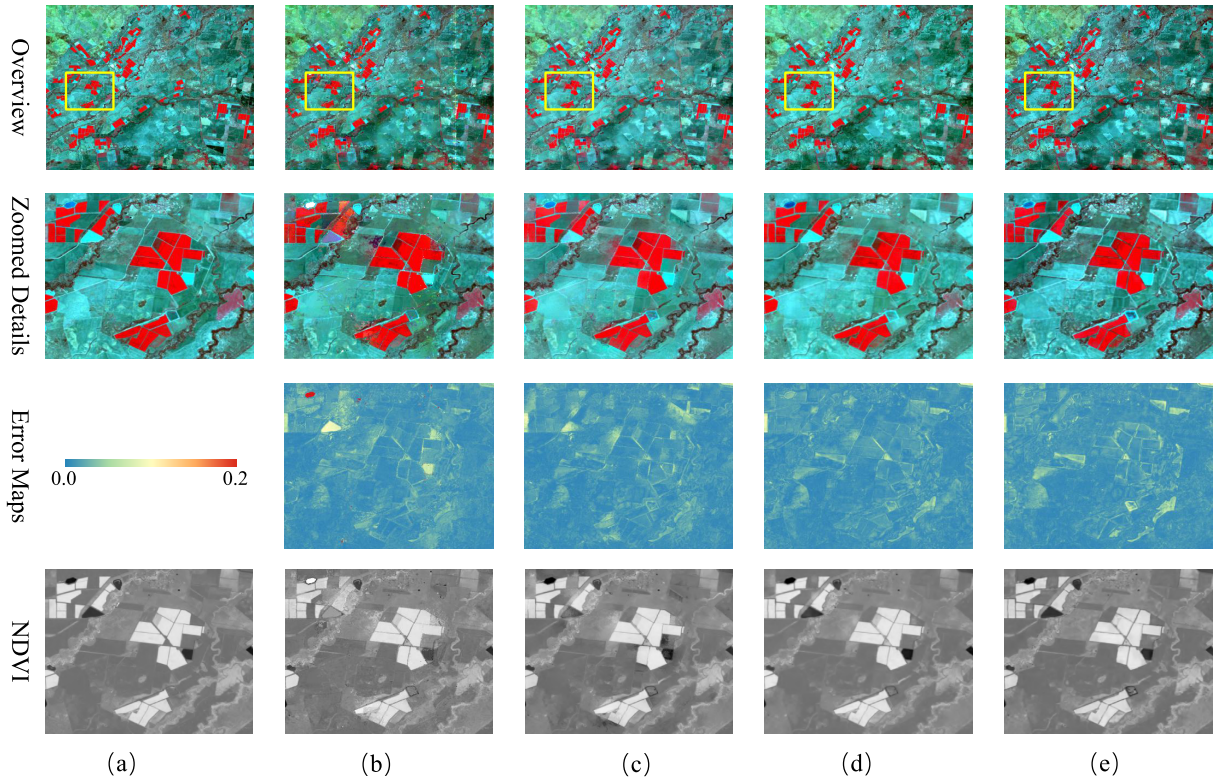


Fig. 9. Fusion results for the LGC area on March 2, 2005 with different spatiotemporal fusion methods (the last three rows correspond to the region in the first row marked with yellow rectangles). (a) Ground Truth. (b) STARFM. (c) FSDAF. (d) EDCSTFN. (e) GAN-STFM.

too much noise from the observation of the first row, while the other three can produce satisfactory results. The second row corresponds to the region marked with yellow rectangles in the first row. The third is the error map by comparing it with the ground truth. The value range is stretched between 0.1 and 0.2 to highlight the errors. Models except STARFM have

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TABLE VII

QUANTITATIVE EVALUATION RESULTS FOR THE CIA AREA ON NOVEMBER 9, 2001 WITH DIFFERENT REFERENCES

Model	Reference Date	MAE	RMSE	SAM	SSIM
STARFM	Oct. 17th	0.0266	0.0784	0.4210	0.7735
	Nov. 2nd	0.0224	0.0808	0.3704	0.8261
	Nov. 25th	0.0285	0.1091	0.6767	0.7848
	Dec.4th	0.0373	0.1344	0.7677	0.7470
FSDAF	Oct. 17th	0.0230	0.0405	0.2582	0.8024
	Nov. 2nd	0.0177	0.0321	0.2183	0.8639
	Nov. 25th	0.0197	0.0355	0.2414	0.8202
	Dec.4th	0.0229	0.0395	0.2641	0.8054
EDCSTFN	Oct. 17th	0.0236	0.0389	0.2491	0.8213
	Nov. 2nd	0.0159	0.0290	0.1949	0.8768
	Nov. 25th	0.0214	0.0351	0.2201	0.8395
	Dec.4th	0.0297	0.0426	0.2400	0.8119
GAN-STFM	Oct. 17th	0.0185	0.0334	0.2183	0.8461
	Nov. 2nd	0.0155	0.0284	0.1921	0.8801
	Nov. 25th	0.0163	0.0307	0.2049	0.8571
	Dec.4th	0.0163	0.0305	0.2036	0.8660

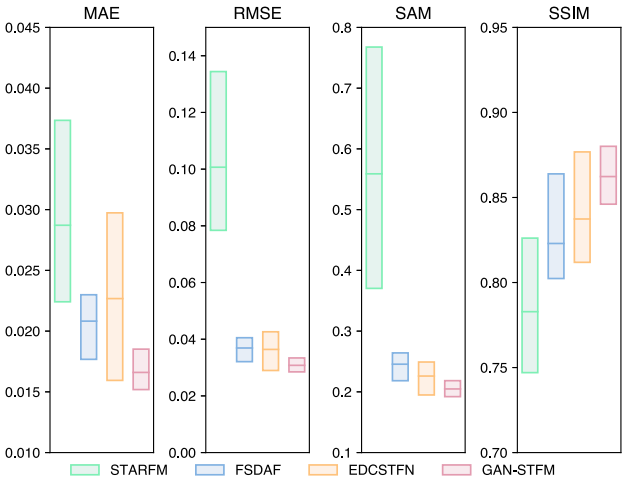
TABLE VIII

QUANTITATIVE EVALUATION RESULTS FOR THE LGC AREA ON MARCH 2, 2005 WITH DIFFERENT REFERENCES

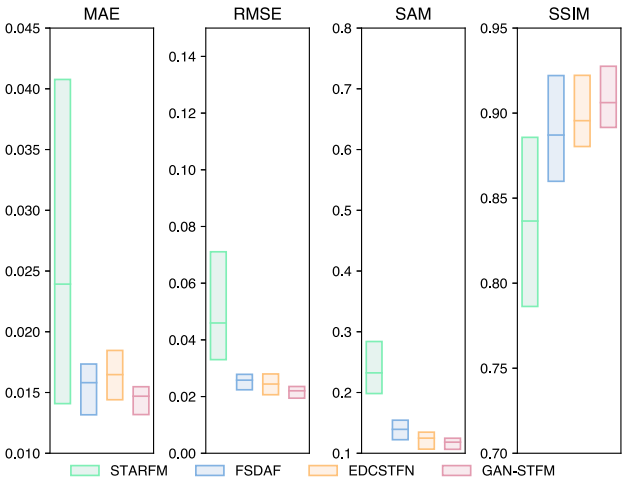
Model	Reference Date	MAE	RMSE	SAM	SSIM
STARFM	Jan. 13th	0.0408	0.0711	0.2840	0.7863
	Jan. 29th	0.0169	0.0338	0.2147	0.8376
	Feb. 4th	0.0141	0.0330	0.1983	0.8857
FSDAF	Jan. 13th	0.0169	0.0271	0.1414	0.8793
	Jan. 29th	0.0173	0.0278	0.1544	0.8599
	Feb. 14th	0.0132	0.0224	0.1223	0.9221
EDCSTFN	Jan. 13th	0.0185	0.0280	0.1332	0.8841
	Jan. 29th	0.0165	0.0246	0.1347	0.8804
	Feb. 14th	0.0144	0.0207	0.1068	0.9222
GAN-STFM	Jan. 13th	0.0154	0.0235	0.1235	0.8994
	Jan. 29th	0.0154	0.0231	0.1247	0.8916
	Feb. 14th	0.0131	0.0194	0.1065	0.9275

similar error distribution. The last is the calculated normalized difference vegetation index (NDVI), and the GAN-STFM produces the closest approximation. The prediction results for the LGC area on March 2, 2005 are shown in Fig. 9. All the models can produce satisfactory results visually from the first two rows. The DCSTFN and GAN-STFM have the lowest error and their NDVIs resemble the ground truth most from the third and fourth rows. The STARFM still has the highest error. Considering that the STARFM is one of the earliest spatiotemporal methods, it is understandable that the robustness of the algorithm is not very high. Generally, from both the quantitative evaluation and visual inspection, the GAN-STFM model can compete with the existing fusion models with fewer inputs and less restriction.

3) Evaluation on References: The selection of reference images for the existing spatiotemporal fusion models is critical to the prediction accuracy. Nearly, all the existing fusion models require the acquisition date of reference image pairs which is close to the prediction date. The following experiment aims to explore the impact of reference images for fusion models by using different references to perform the prediction on the same date. Table VII shows the quantitative evaluation results



(a)



(b)

Fig. 10. Quantitative evaluation for the different models using different reference images. (a) Evaluation results for the CIA area on November 9, 2001. (b) Evaluation results for the LGC area on March 2, 2005.

for the CIA area on November 9, 2001 with the four different reference images on October 17, November 2, November 25, and December 4. Table VIII lists the evaluation results for the LGC area on March 2, 2005 with the three different reference images on January 13, January 29, and February 4. The metrics observed from the tables show a slight difference among the comparable groups for the GAN-STFM model, while the performance of the other three models varies significantly. The evaluation results are visualized in Fig. 10 to reconfirm the conclusion intuitively. The performance of the four fusion models is distinguished with different colors. The top and bottom of the boxes denote the maximum and minimum values among the test groups with different references, respectively. The vertical lines that go through the boxes are the mean values for each test group. Clearly, the GAN-STFM model can predict with high accuracy consistently with different references, whereas others exhibit certain large fluctuations, especially for the STARFM model. Figs. 11 and 12 show the fusion results of GAN-STFM on CIA and LGC areas,

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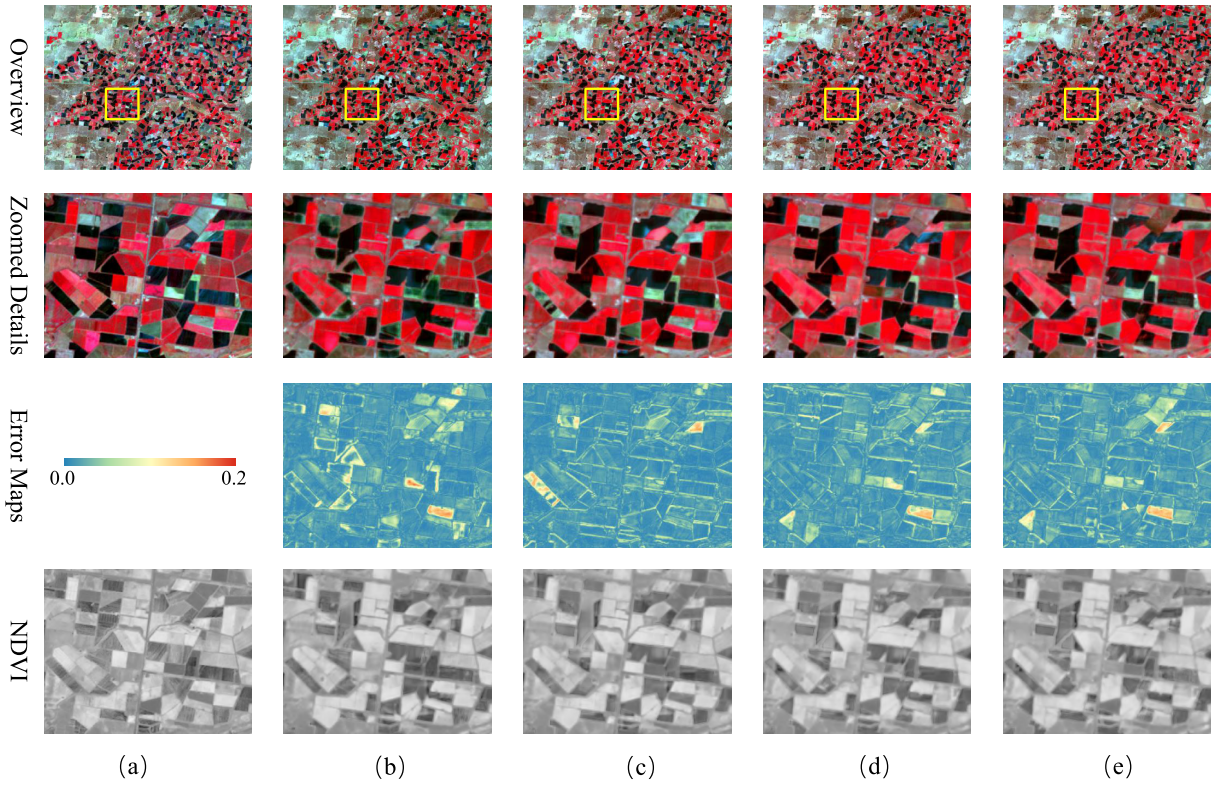


Fig. 11. Fusion results for the CIA area on November 9, 2001 with reference images acquired on different dates (the last three rows correspond to the region in the first row marked with yellow rectangles). (a) Ground Truth. (b) October 17. (c) November 2. (d) November 25. (e) December 4.

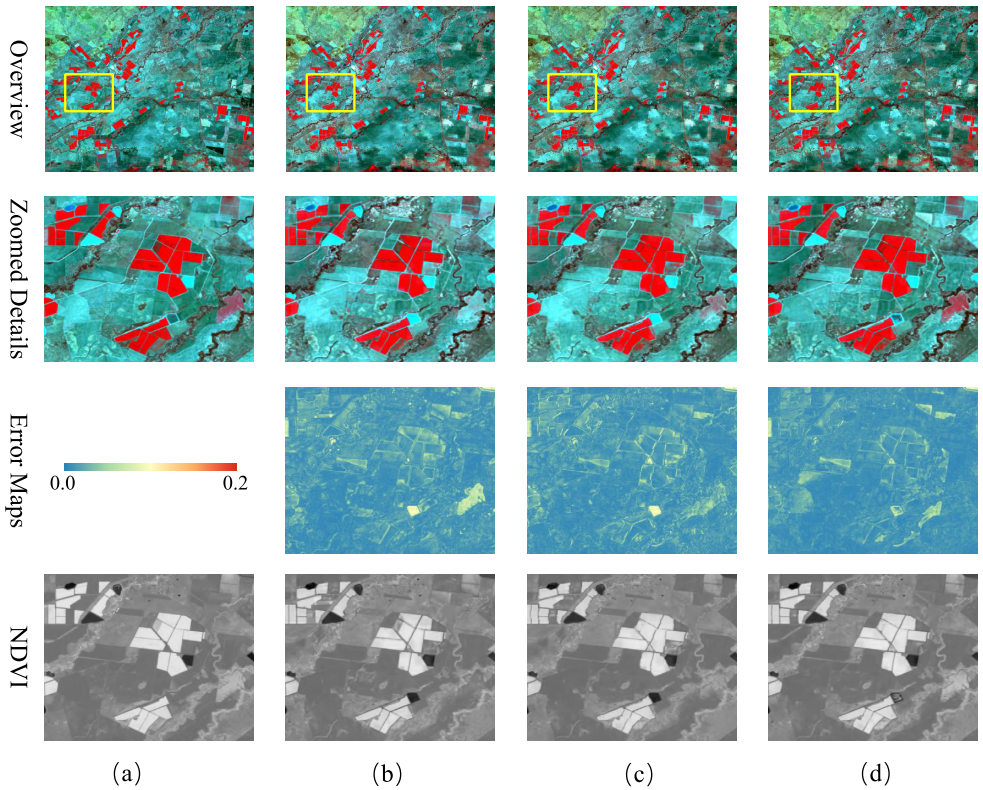


Fig. 12. Fusion results for the LGC area on March 2, 2005 with reference images acquired on different dates (the last three rows correspond to the region in the first row marked with yellow rectangles). (a) Ground Truth. (b) January 13. (c) January 29. (d) February 14.

respectively. The prediction with reference of October 17 on the CIA area and the prediction with reference January 13 on the LGC area produce the highest errors, but overall the generated images, error maps, and NDVI look very similar visually. Thus, it can safely conclude that the GAN-STFM can be used to do spatiotemporal fusion with an arbitrary fine-resolution reference image without rigid time constraints like traditional methods.

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IV. Conclusion

Currently, the existing spatiotemporal fusion models almost require at least three images as inputs, and rigid time constraints are applied to reference image selection to guarantee fusion accuracy. To improve the usability of the fusion model, a CGAN-based fusion model termed GAN-STFM is proposed to reduce the model inputs and lessen the restrictions on model inputs with the normalization technique. Experiments demonstrate, from both the quantitative evaluation and visual inspection, that the proposed GAN-STFM model can compete with the existing fusion models. Little influence is observed on fusion results when putting different reference images for the GAN-STFM model. With fewer inputs and less restriction, the GAN-STFM shows more flexibility and has a quite promising prospect for practical data production.

After years of research and development, the spatiotemporal fusion accuracy has been gradually improved over time, but one must admit that further improvement is getting rather harder. We believe that in the future, more attention should be paid to the robustness and usability of spatiotemporal fusion models. This article, from one perspective, tries to improve the model usability by proposing the reference-insensitive mechanism to lower the requirements for model inputs. Our future work will mainly devote to design advanced models capable of handling input noises of a certain degree automatically, improving the model robustness.

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